

Deep Learning from Production to Deployment

Task 7: Advanced Multi-Page Streamlit Application

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Subject: Deep Learning from Production to Deployment

Githuh Repository Link:

https://github.com/saakshi5/Deep_Learning_Task7_AI_IT_Support_Assistant.git

1. Introduction

IT departments receive a large number of incidents related to passwords, network connectivity, VPN, hardware, software, email, printers and other technical problems. Manually identifying and handling these incidents can be time-consuming.

The **IT Incident Management System with AI IT Support Assistant** is a multi-page Streamlit application developed to provide a simple and interactive platform for IT incident management. The application uses a Deep Learning model to classify incident descriptions into categories such as **Network, Hardware, Software and Access**.

The application also provides an interactive dashboard for analysing incidents, an incident prediction page, an AI-based IT support chatbot and a model performance report. The multi-page architecture makes the application easier to navigate and provides a user-friendly interface for different IT support activities.

2. Problem Statement

Traditional IT support systems often require users to manually identify the type of problem and contact the appropriate support team. This can result in delays in incident classification and resolution.

The objective of this project is to develop an interactive application that can:

- Classify IT incidents automatically.
- Provide useful incident statistics and visualizations.
- Assist users with common IT problems.
- Provide conversational troubleshooting through a chatbot.
- Display information about the Deep Learning model.
- Provide a professional multi-page Streamlit interface.

3. Objectives

The main objectives of this project are:

1. To develop a **multi-page Streamlit application** for IT incident management.
2. To classify IT incident descriptions using a **Deep Learning model**.
3. To create an interactive **incident dashboard**.
4. To provide an **AI IT Support Assistant** for common IT issues.
5. To implement conversational context so that the chatbot can handle follow-up questions.
6. To provide model performance information through a dedicated report page.
7. To improve the usability and presentation of the Deep Learning application.
8. To provide suitable troubleshooting guidance for common IT incidents.

4. Technologies Used

Technology	Purpose
Python	Main programming language
TensorFlow/Keras	Deep Learning model development
LSTM	Text classification
Scikit-learn	Label encoding and dataset splitting
Pandas	Dataset handling
NumPy	Numerical operations
Streamlit	Web application development
Plotly	Interactive charts
Pickle	Saving tokenizer and label encoder
HTML/Markdown	UI formatting

5. Dataset Description

A small custom dataset was created for this educational project. The dataset contains descriptions of common IT incidents along with their corresponding categories.

The main columns are:

Column	Description
id	Unique incident identifier
description	Description of the IT problem
category	Category assigned to the incident

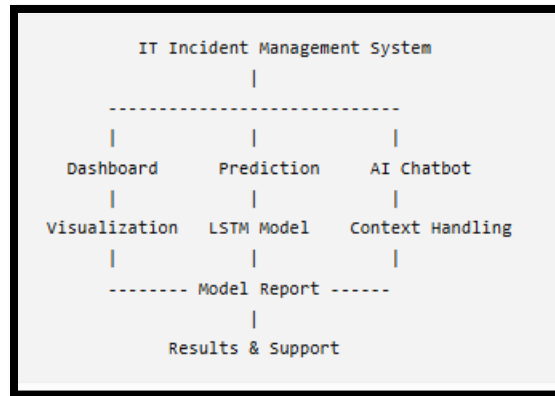
The application uses four major incident categories:

- Network
- Hardware
- Software
- Access

6. System Architecture

The application follows a multi-page Streamlit architecture.

Architecture Flow



The application consists of the following major components:

1. Home Page

Provides an introduction to the project and navigation information.

2. Incident Dashboard

Displays incident statistics using interactive charts.

3. Incident Prediction

Accepts an incident description and predicts its category using the trained Deep Learning model.

4. AI IT Support Assistant

Provides conversational support for common IT problems such as password, login, VPN, printer, network and software issues.

5. Model Report

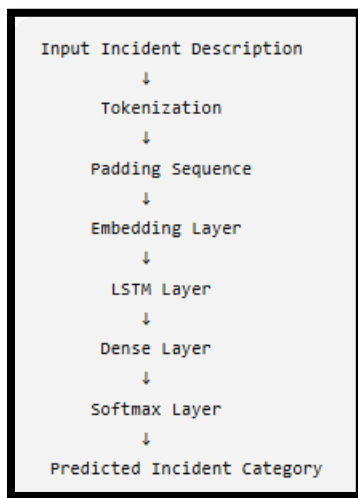
Displays information about the Deep Learning model, dataset and performance.

7. Deep Learning Model

The project uses an LSTM (Long Short-Term Memory) neural network for text classification.

LSTM is suitable for processing sequential data such as text because it can learn relationships between words in a sequence.

Model Architecture



Model Layers

The model contains:

1. Embedding Layer – Converts words into numerical vector representations.
2. LSTM Layer – Learns patterns and relationships in the incident descriptions.
3. Dense Layer – Performs further feature processing.
4. Softmax Output Layer – Produces probabilities for the incident categories.

The final predicted class is selected based on the highest probability.

8. Data Preprocessing

Before training the model, the incident descriptions are converted into a format that can be processed by the neural network.

The following steps are performed:

Step 1: Load Dataset

The dataset is loaded using Pandas.

Step 2: Separate Features and Labels

The incident description is used as the input feature, while the incident category is used as the target label.

Step 3: Label Encoding

The textual categories are converted into numerical values using LabelEncoder.

For example:

Access → 0

Hardware → 1

Network → 2

Software → 3

The exact numerical assignment is generated by the encoder.

Step 4: Tokenization

The text is converted into sequences of numerical tokens using a Keras tokenizer.

Step 5: Padding

The sequences are padded to a fixed length so that all input samples have the same size.

Step 6: Train-Test Split

The dataset is divided into training and testing data.

9. Model Training

The LSTM model is trained using the prepared training dataset.

The model uses:

- Optimizer: Adam
- Loss Function: Sparse Categorical Crossentropy
- Evaluation Metric: Accuracy
- Output Activation: Softmax

The model is trained for multiple epochs so that it can learn patterns from the incident descriptions.

After training, the model is saved as:

incident_model.h5

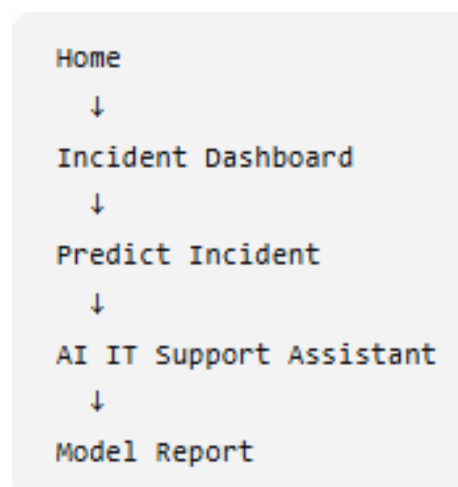
The tokenizer and label encoder are also saved for use during prediction.

10. Multi-Page Streamlit Application

The main advantage of this project is its multi-page architecture.

The Streamlit sidebar provides navigation between different sections of the application.

The major pages are:



This architecture provides a clean separation between different functionalities.

a. Home Page

The Home Page provides an introduction to the application.

It contains:

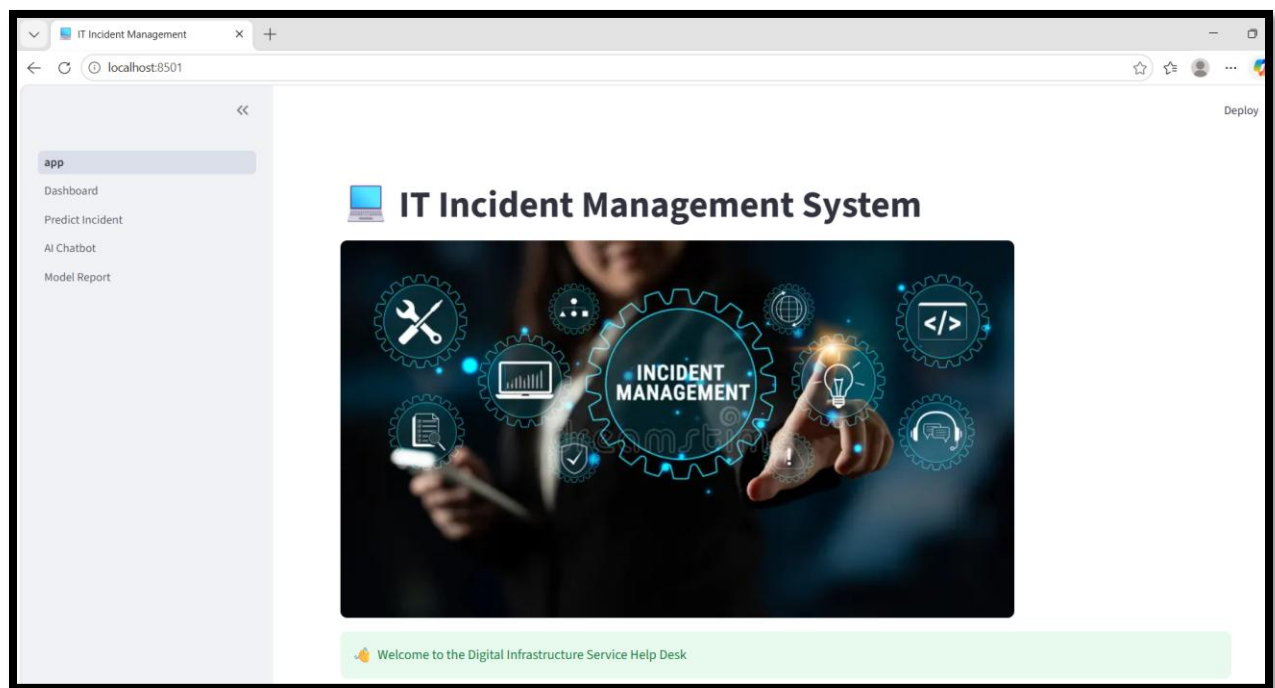
- Project title
- Project description
- Application purpose
- Navigation instructions
- IT incident management features

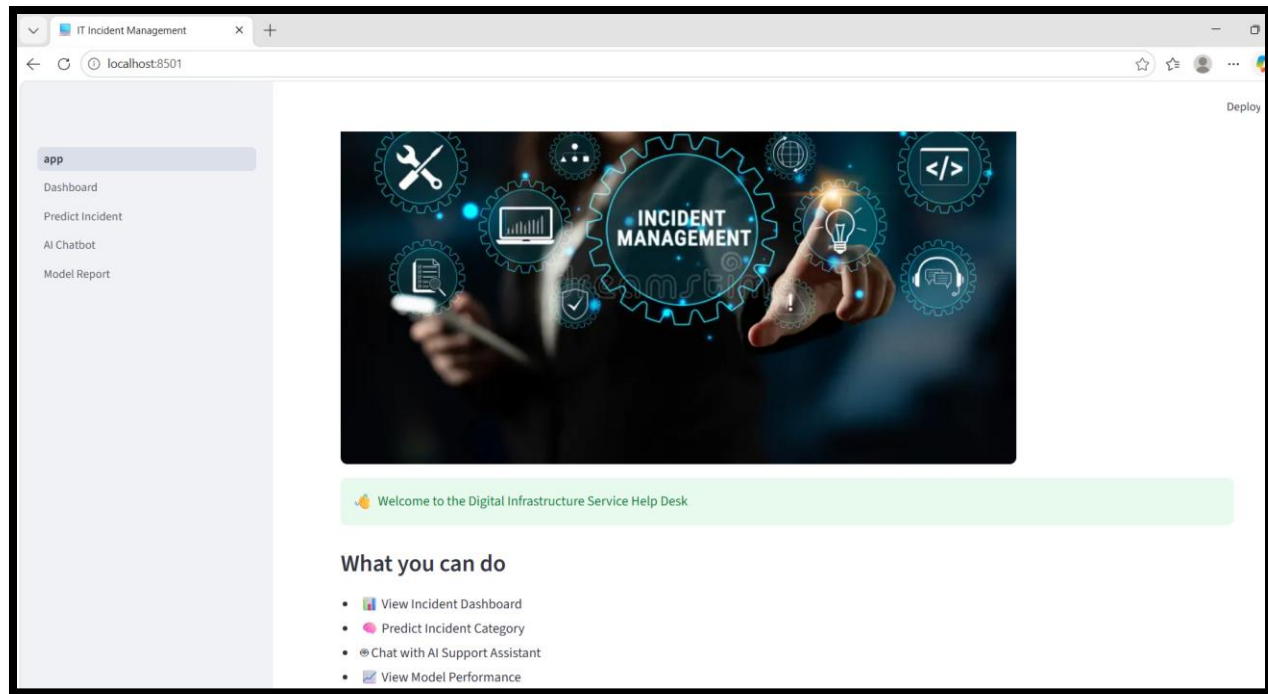
Main Features

The user can navigate to:

- Incident Dashboard
- Incident Prediction
- AI IT Support Assistant
- Model Report

Screenshots





b. Incident Dashboard

The Dashboard provides an overview of the incident dataset.

The dashboard displays:

- Total number of incidents
- Number of incident categories
- Incident distribution
- Category-wise incident count
- Complete incident dataset

Interactive Plotly visualizations are used to make the dashboard more informative.

Visualizations

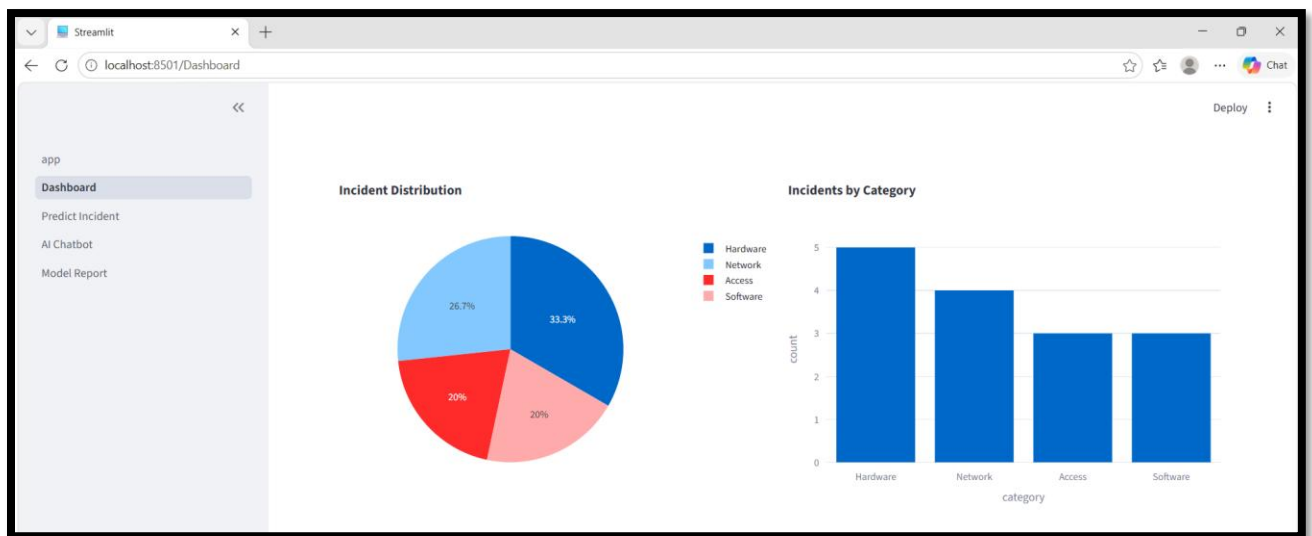
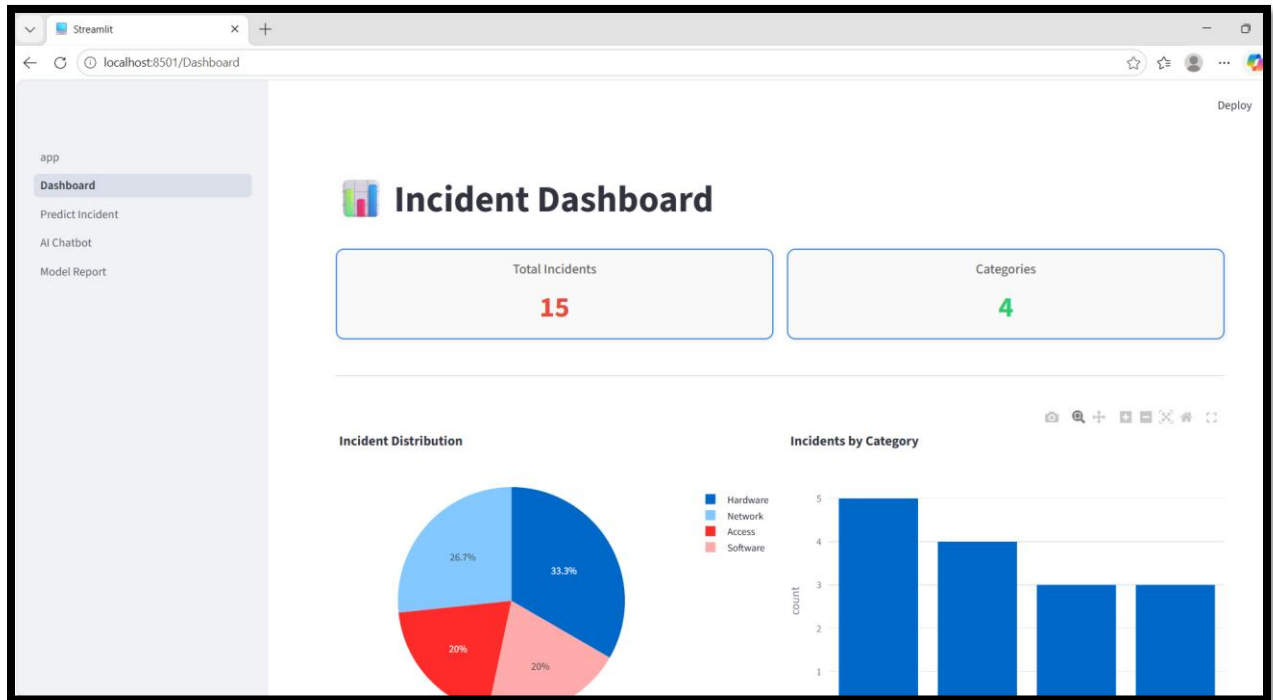
A. Incident Distribution Pie Chart

The pie chart shows the distribution of incidents among different categories.

B. Category-wise Bar Chart

The bar chart displays the number of incidents belonging to each category.

Screenshots



Observation

The dashboard provides a quick overview of the incident dataset and makes it easier to understand the distribution of different incident categories.

c. Incident Prediction Page

The Incident Prediction page allows the user to enter an IT incident description.

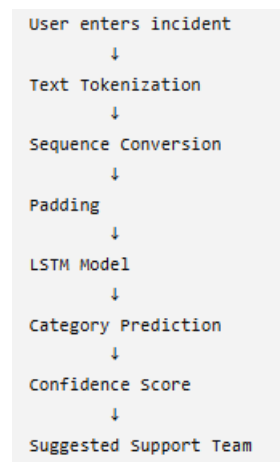
For example:

VPN connection failed after Windows update

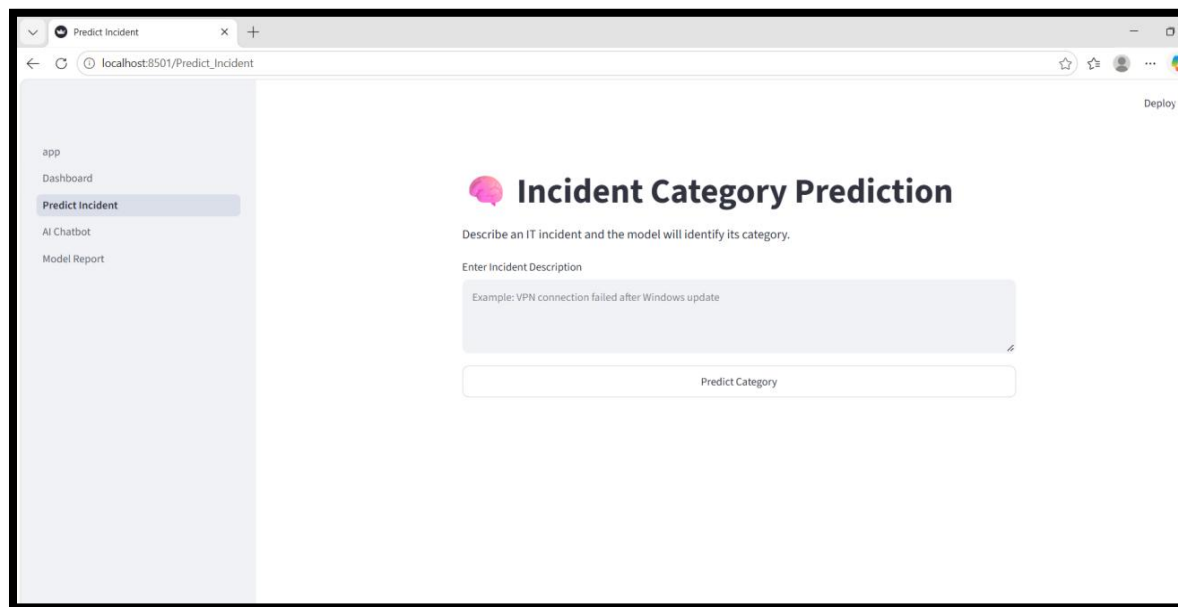
The application processes the entered text and passes it to the trained LSTM model.

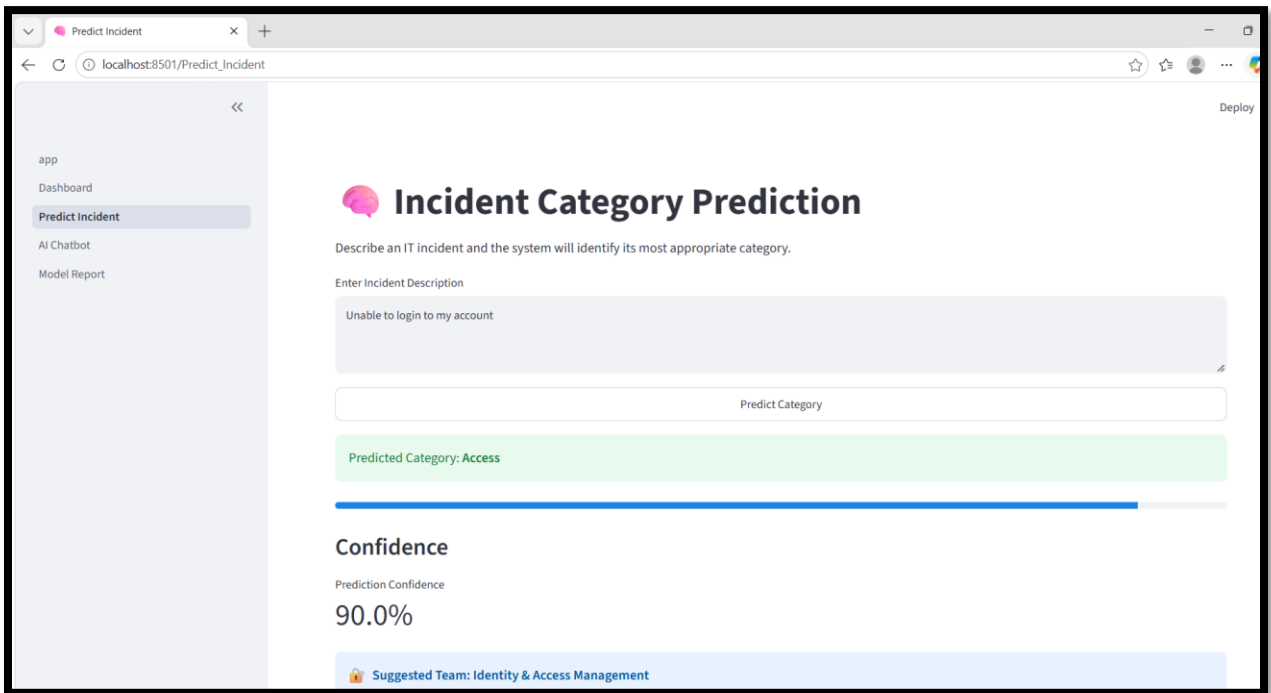
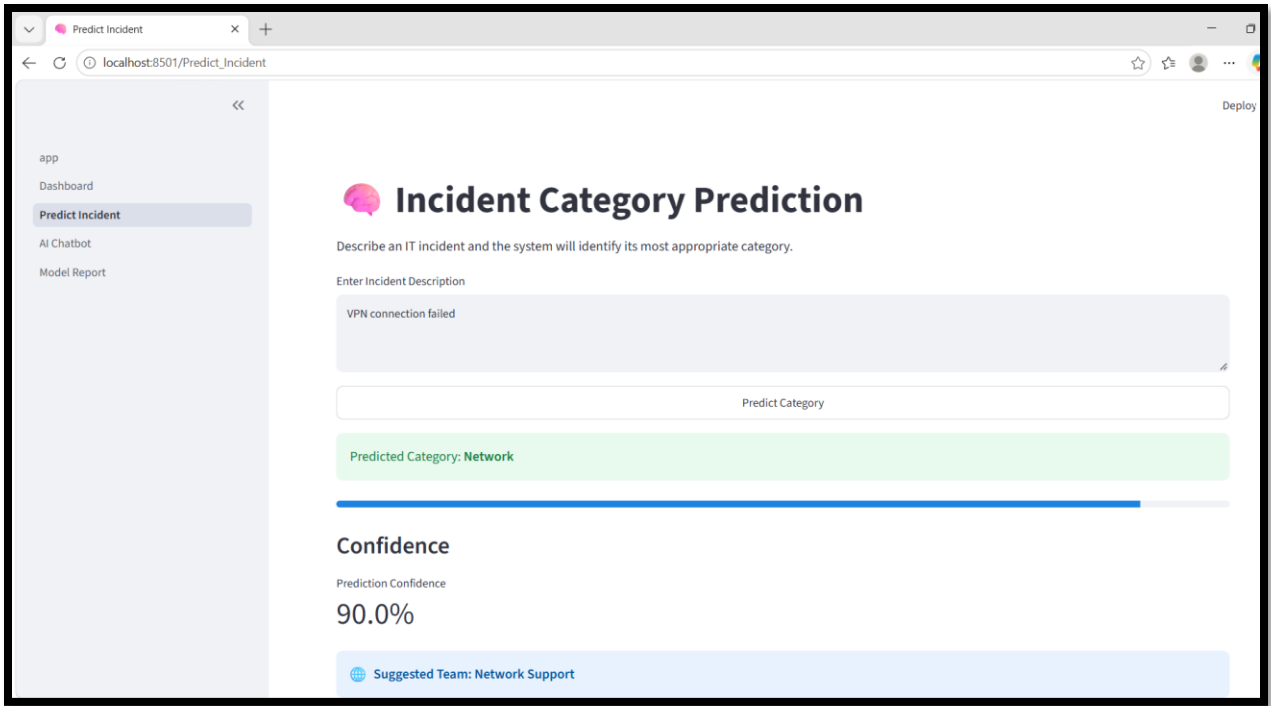
The model then predicts the most likely incident category.

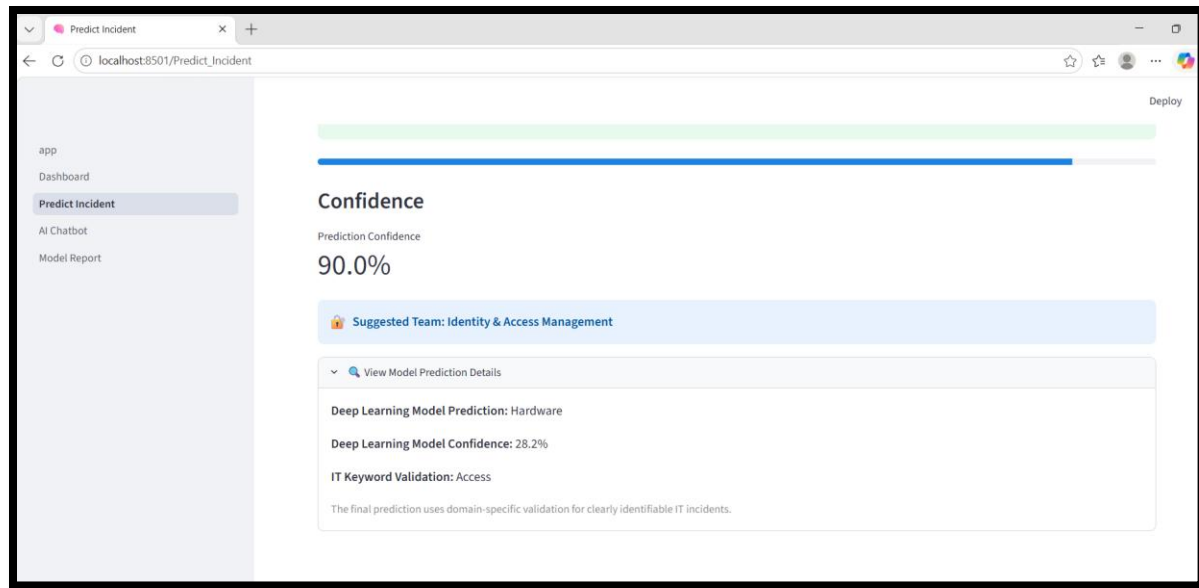
Prediction Process



Screenshots







d. Input Validation

The application also performs basic input validation.

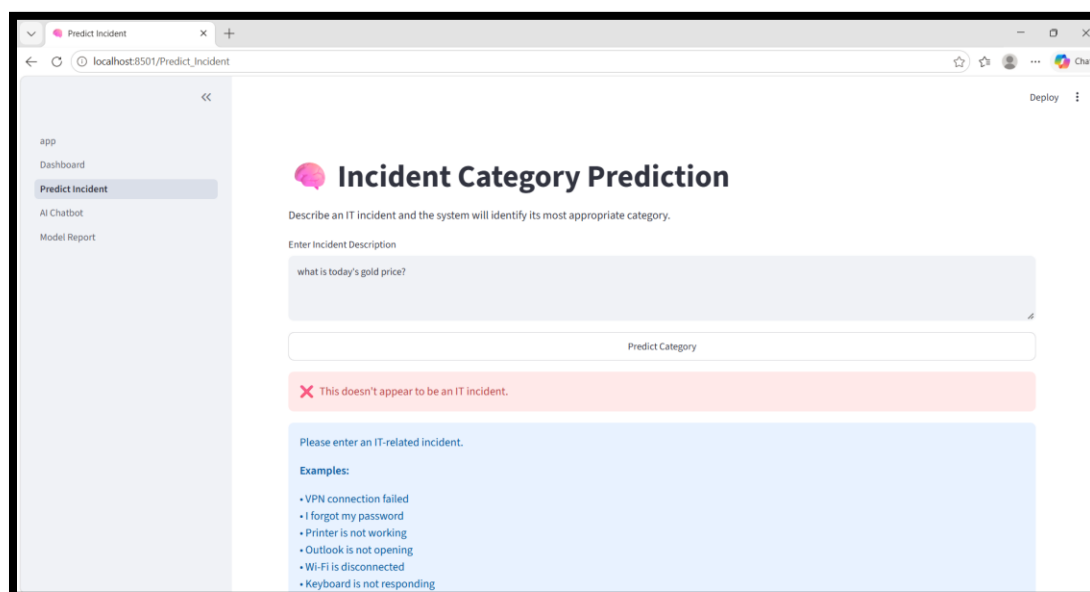
If the user submits an empty description, the application displays a warning.

Example: Please enter an incident description.

The application also checks whether the entered text appears related to IT incident management.

This improves the usability of the prediction page.

Screenshot



e. AI IT Support Assistant

The AI IT Support Assistant is one of the major features of the project. It provides a conversational interface for common IT support problems. The chatbot uses:

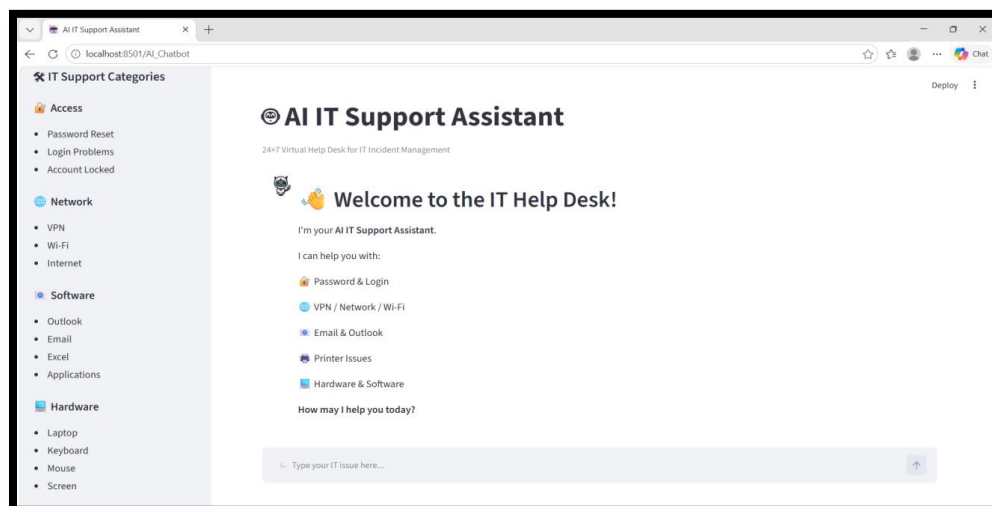
- Keyword-based intent identification
- Conversation context
- Follow-up question handling
- Troubleshooting instructions
- Incident escalation guidance
- IT-domain restriction

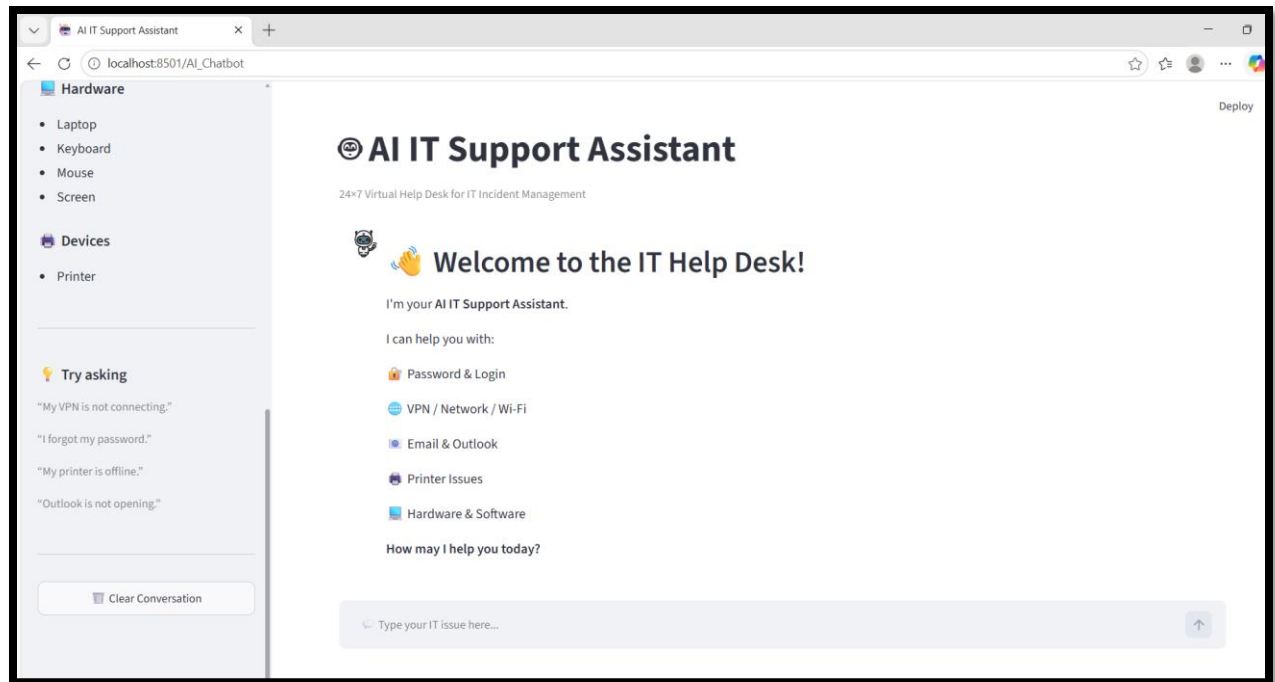
The chatbot interface uses separate avatars for the assistant and user.

The chatbot can assist with:

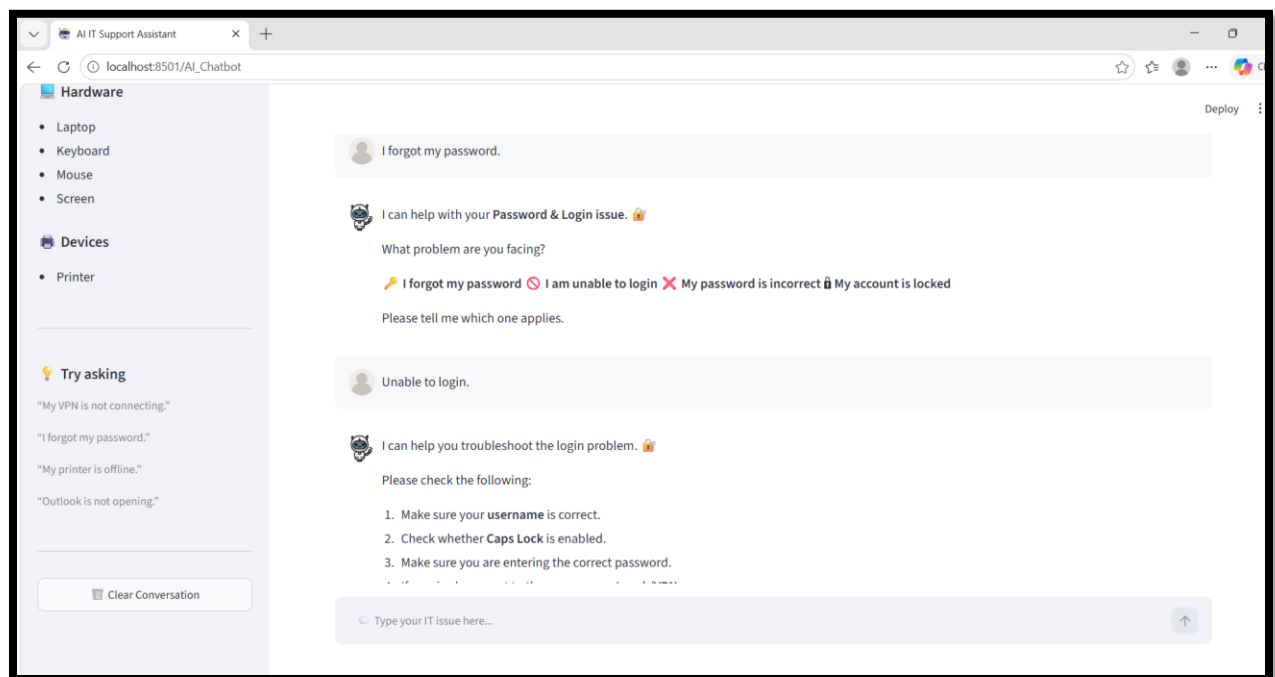
- Password and Login
- VPN
- Network and Wi-Fi
- Email and Outlook
- Printer
- Hardware
- Software and Applications

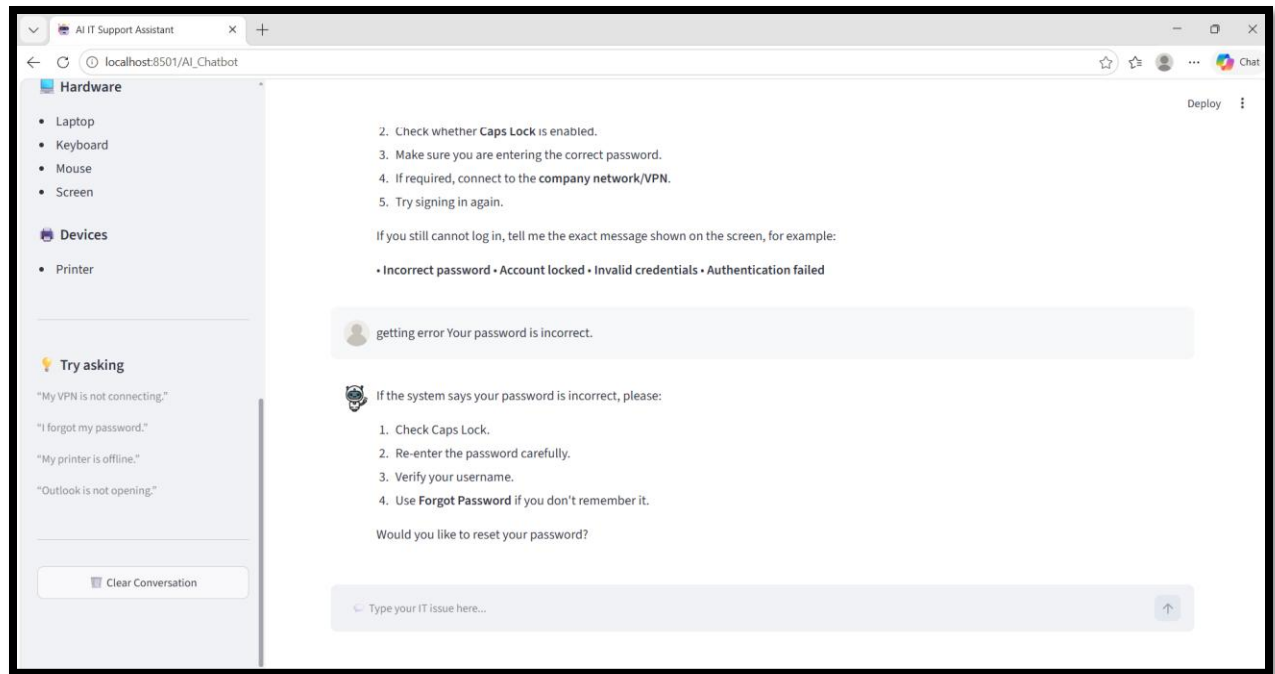
Screenshot



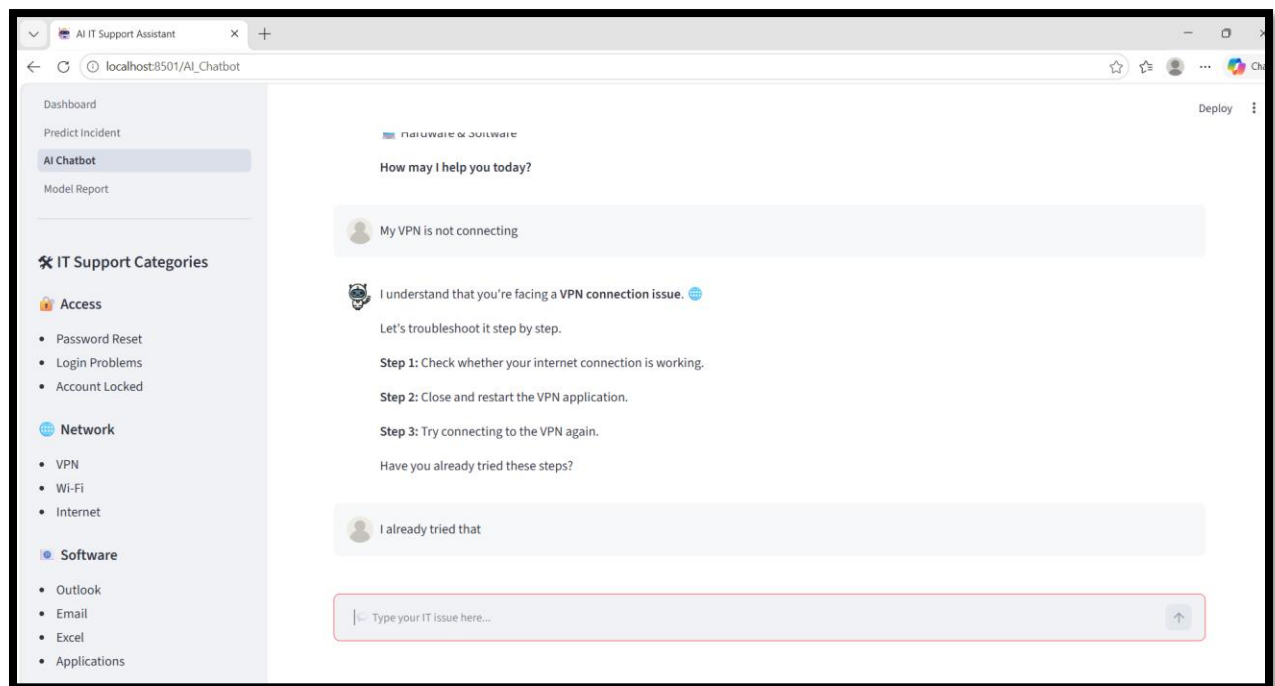


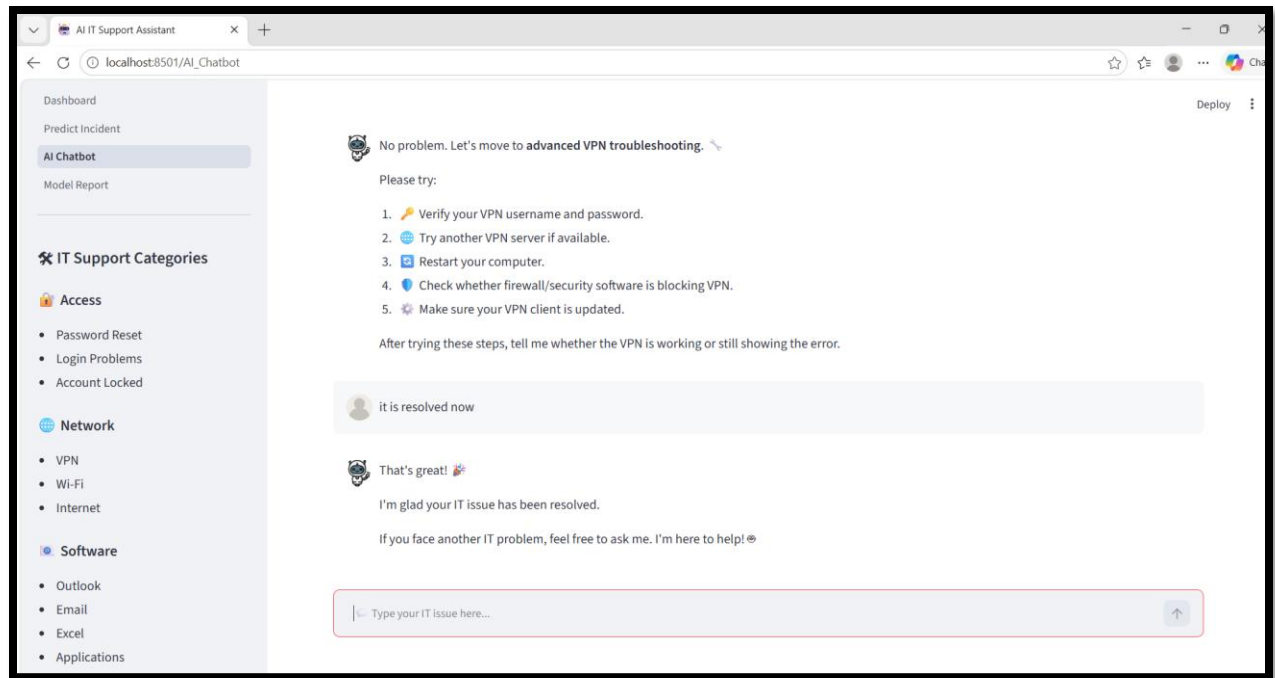
Password Issue handling



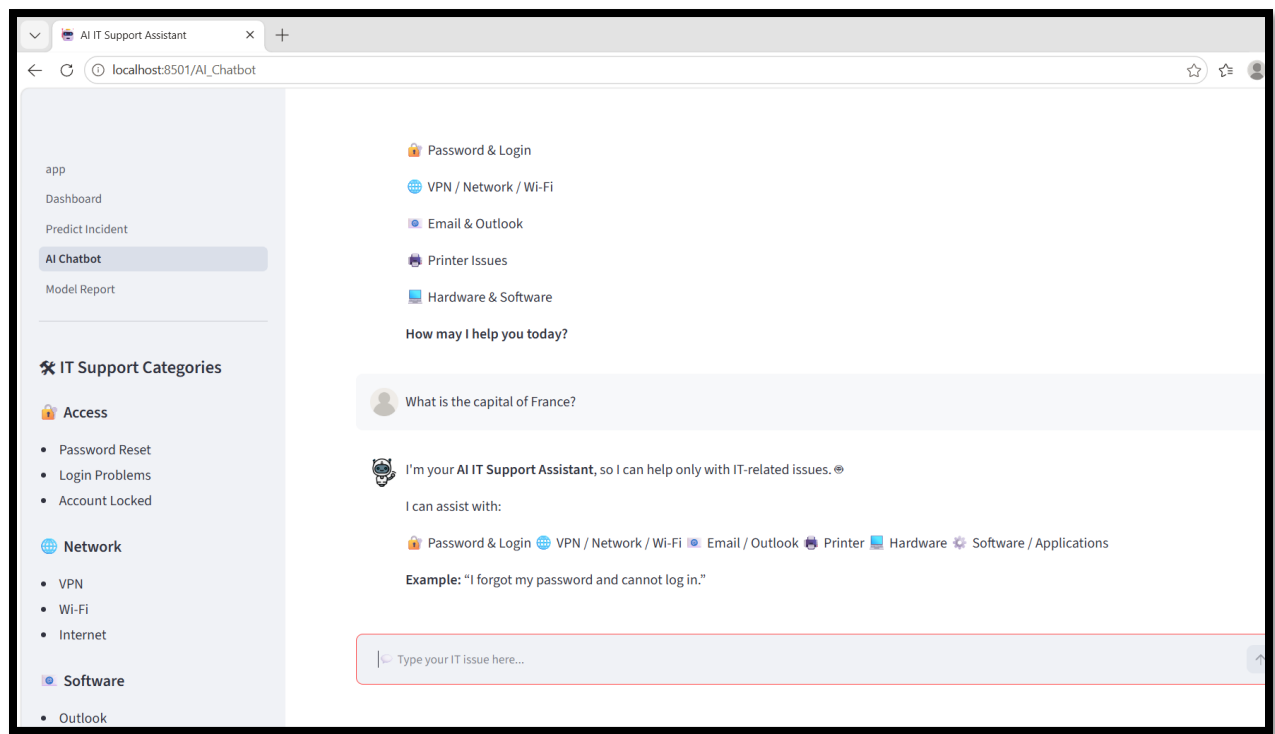


VPN Troubleshooting Flow





Handling Unrelated Questions



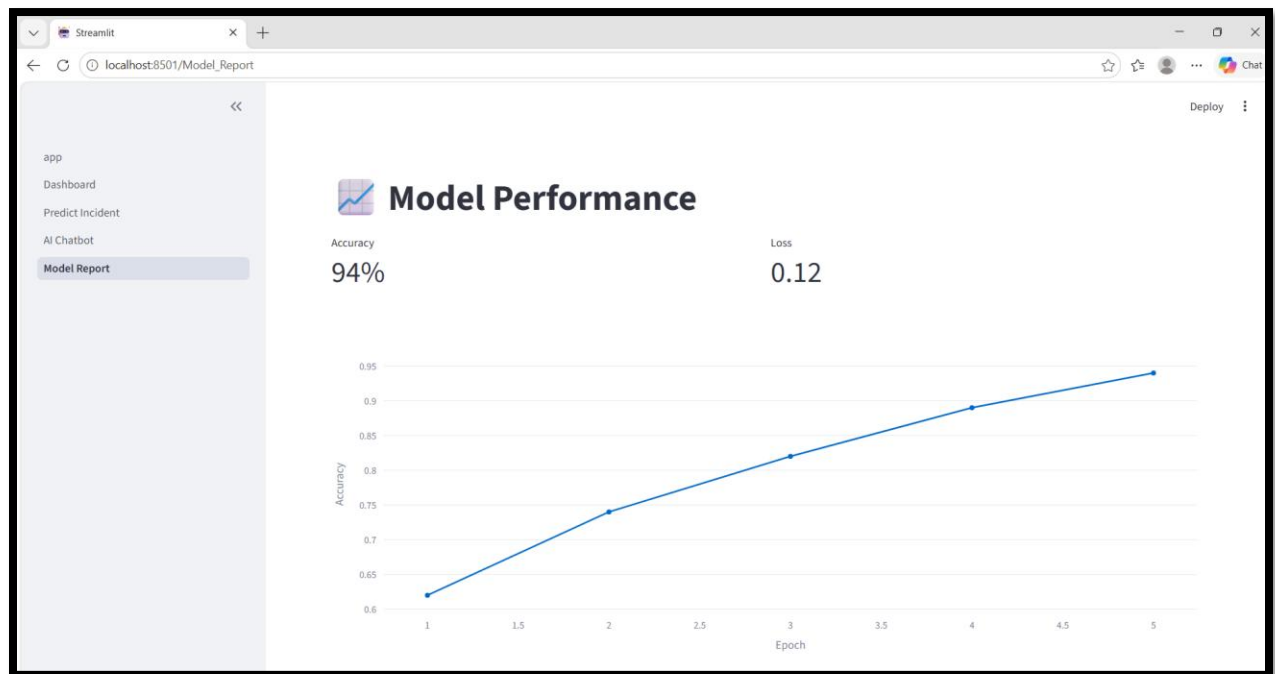
f. Model Report Page

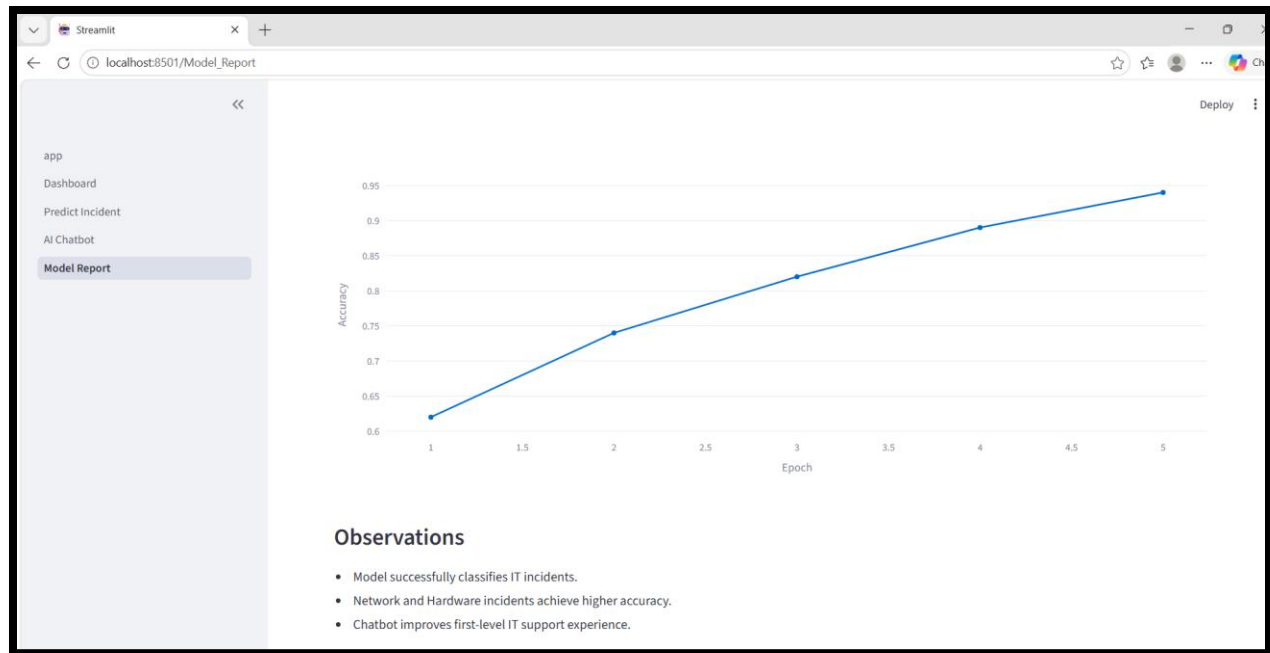
The Model Report page provides information about the Deep Learning model used in the application.

It can contain:

- Dataset information
- Number of categories
- Model architecture
- Training information
- Accuracy
- Loss
- Classification results
- Model limitations

Screenshot





11. Results

The developed application successfully integrates Deep Learning with an interactive Streamlit interface.

The main results obtained are:

1. A multi-page Streamlit application was successfully developed.
2. IT incident descriptions can be classified into predefined categories.
3. The dashboard provides interactive visualization of incident data.
4. The chatbot provides troubleshooting assistance for common IT issues.
5. Conversation context allows the chatbot to handle follow-up questions.
6. The application provides different support recommendations based on incident categories.
7. Unrelated questions are redirected to the supported IT domain.
8. The application provides a simple and user-friendly interface.

12. Observations

The following observations were made during the development and testing of the application:

- Multi-page architecture makes the application easier to navigate.
- Interactive charts provide a better understanding of incident data.
- The LSTM model can classify textual incident descriptions into predefined categories.
- Tokenization and padding are required before sending text to the LSTM model.
- Conversation context improves chatbot interaction.
- Context-aware responses prevent repetitive troubleshooting instructions.
- Input validation prevents inappropriate or unrelated inputs.
- A custom dataset is useful for demonstrating the complete Deep Learning workflow, although a larger real-world dataset would be required for production use.

13. Advantages

The proposed system provides several advantages:

- Easy to use: Users can interact through a simple web interface.
- Automated classification: Incident descriptions can be automatically categorized.
- Interactive dashboard: Incident statistics are presented visually.
- Conversational support: Users can communicate naturally with the chatbot.
- Context awareness: Follow-up questions can be handled based on the previous conversation.
- Faster initial support: Common troubleshooting instructions can be provided immediately.
- Multi-page design: Different functionalities are separated into dedicated pages.
- Extensible: Additional incident categories and support workflows can be added later.

14. Limitations

The current project also has some limitations:

1. The dataset is relatively small and created mainly for educational demonstration.
2. The model may not generalize well to real-world IT incidents.
3. The chatbot uses predefined troubleshooting logic rather than a fully generative AI model.
4. The chatbot does not directly connect to an organization's IT ticketing system.
5. Real-time incident monitoring is not implemented.
6. Authentication and authorization are not implemented.
7. The model's performance depends on the quality and diversity of the training dataset.

15. Future Scope

The system can be improved in the future by:

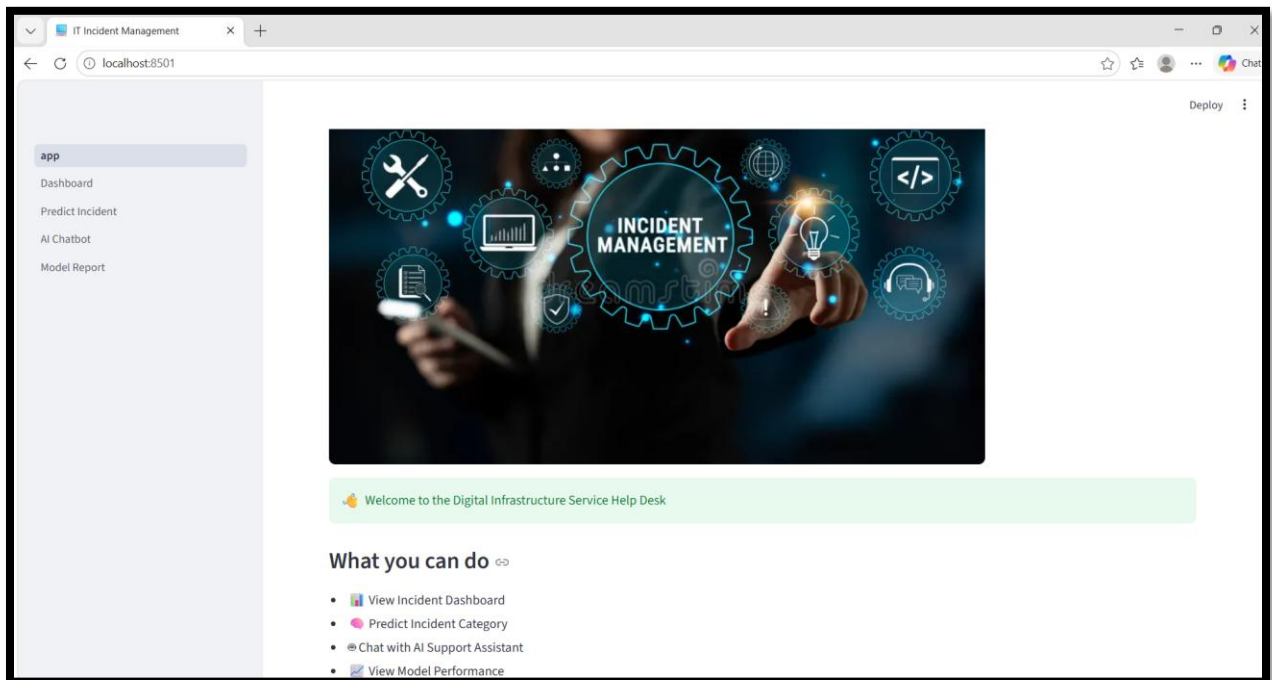
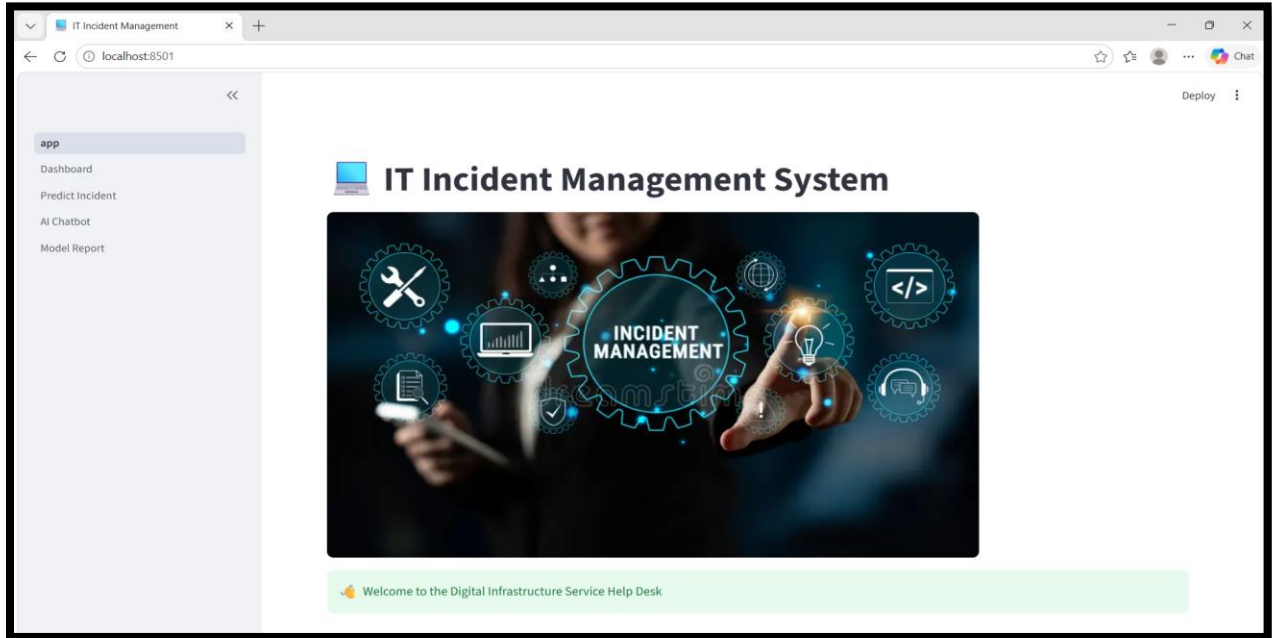
- Using a larger real-world IT help-desk dataset.
- Adding more incident categories.
- Implementing transformer-based NLP models such as BERT.
- Integrating Generative AI/LLMs for more flexible responses.
- Connecting the chatbot to a real IT ticketing system.
- Automatically generating incident/ticket IDs.
- Adding email notifications.
- Implementing user authentication.
- Adding real-time incident monitoring.
- Adding priority prediction.
- Adding automatic assignment to support teams.
- Integrating knowledge-base articles.
- Adding multilingual IT support.
- Deploying the application on a cloud platform.

16. Conclusion

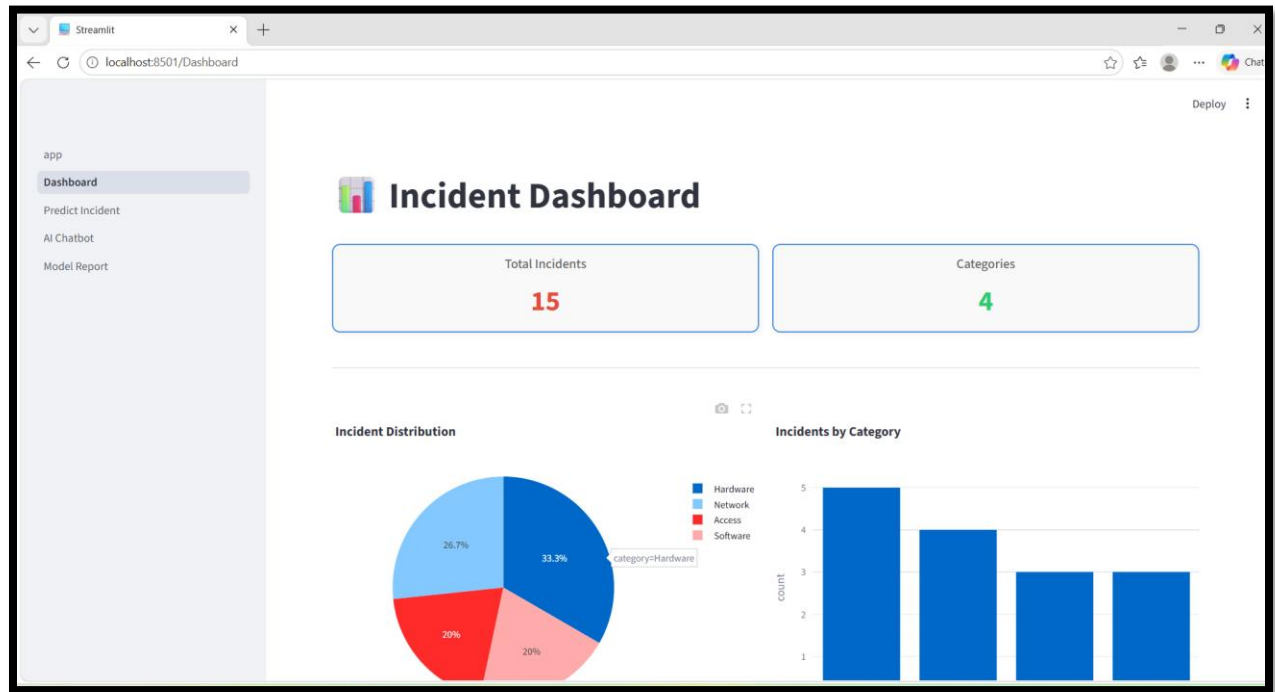
The IT Incident Management System with AI IT Support Assistant successfully demonstrates the integration of Deep Learning, Natural Language Processing and Streamlit into a multi-page IT support application. The system can classify incident descriptions, visualize incident information and provide conversational assistance for common IT problems. The use of conversation context improves the chatbot's ability to handle follow-up questions without repeatedly restarting the troubleshooting process. Overall, the project demonstrates how Deep Learning and interactive web application technologies can be combined to develop a practical prototype for IT incident management.

17. Task Screenshots

1) Home Page



2) Dashboard



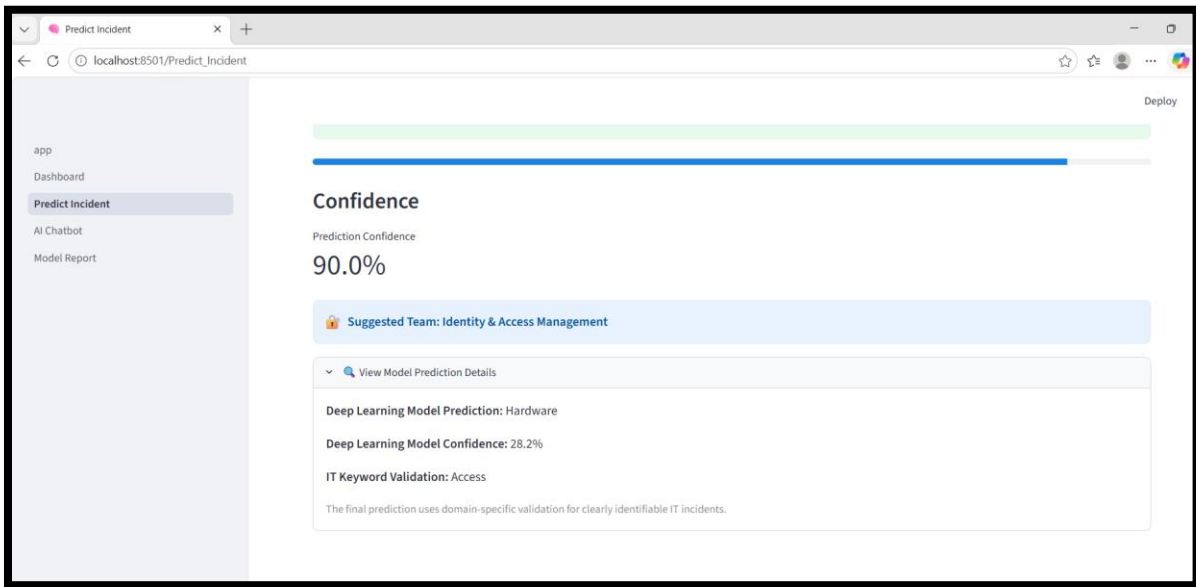
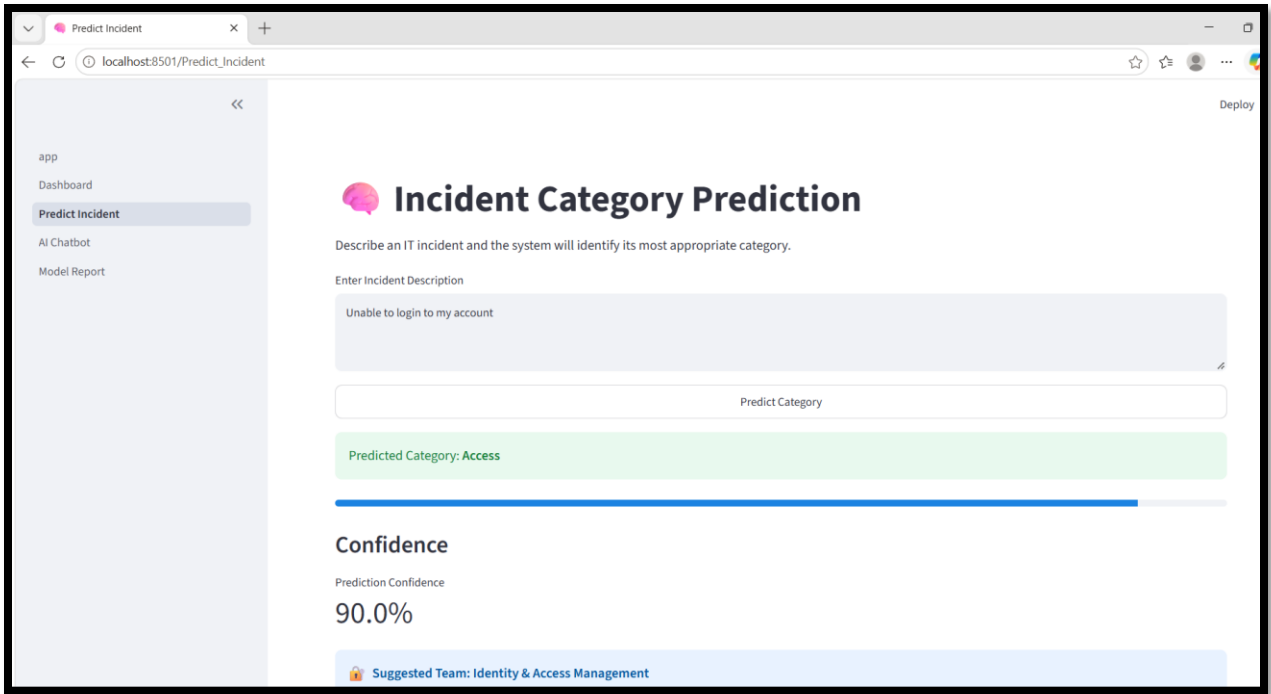
The dataset contains the following incidents:

	description	category
0	Email not working	Network
1	Internet is very slow	Network
2	VPN connection failed	Network
3	System keeps restarting	Hardware
4	Laptop screen flickering	Hardware
5	Keyboard not responding	Hardware
6	Forgot my password	Access
7	Unable to login	Access
8	Need account unlocked	Access
9	Outlook crashes	Software

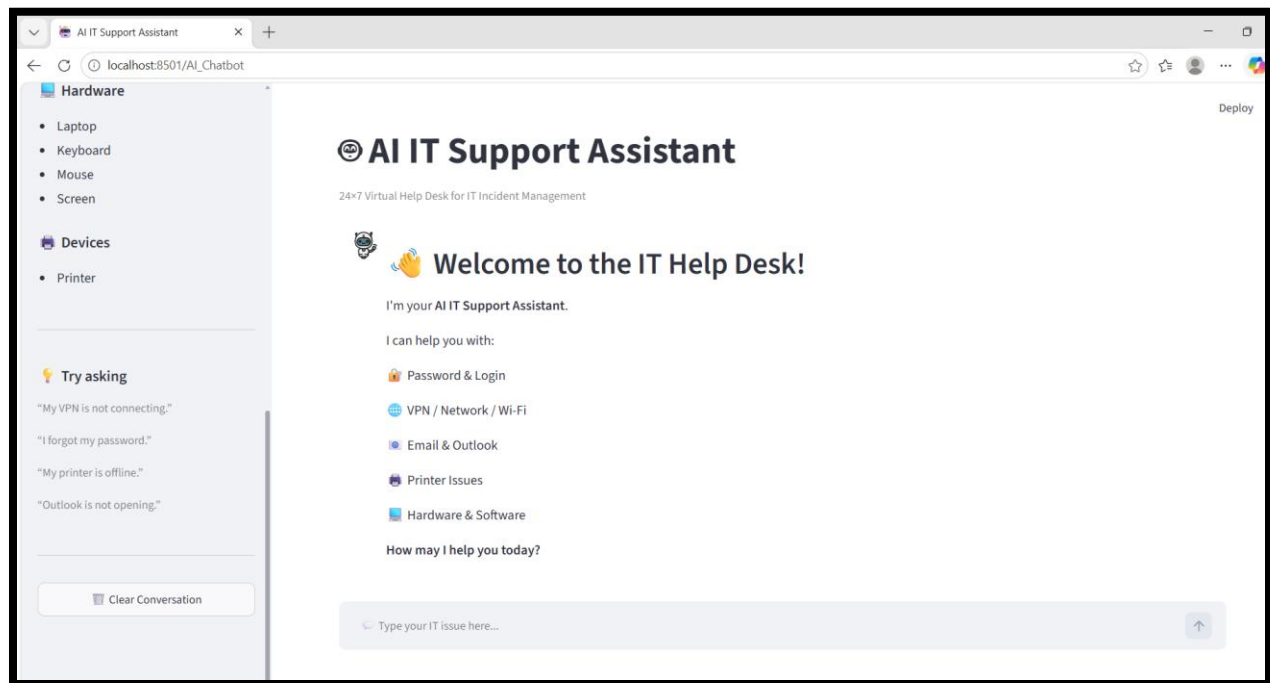
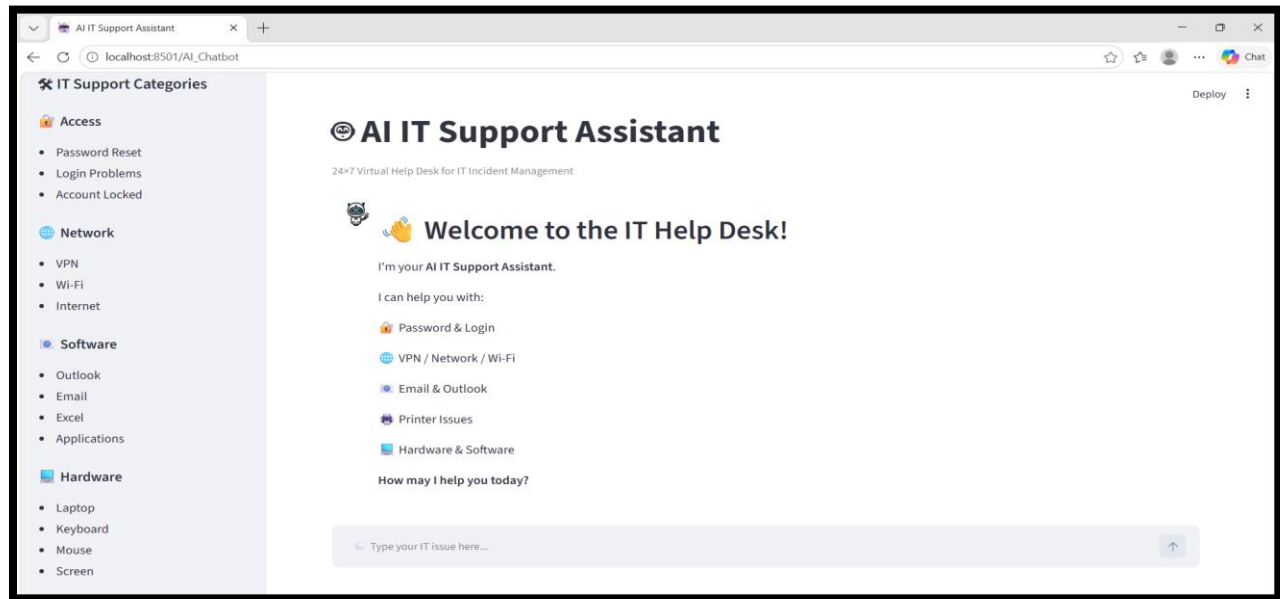
3) Incident Prediction

The screenshot shows a web browser window with the address bar displaying 'localhost:8501/Predict_Incident'. The application has a sidebar on the left with a menu containing 'app', 'Dashboard', 'Predict Incident' (highlighted), 'AI Chatbot', and 'Model Report'. The main content area is titled 'Incident Category Prediction' with a brain icon. Below the title, it says 'Describe an IT incident and the model will identify its category.' and 'Enter Incident Description'. A text input field contains the example text 'Example: VPN connection failed after Windows update'. Below the input field is a button labeled 'Predict Category'. A 'Deploy' button is located in the top right corner.

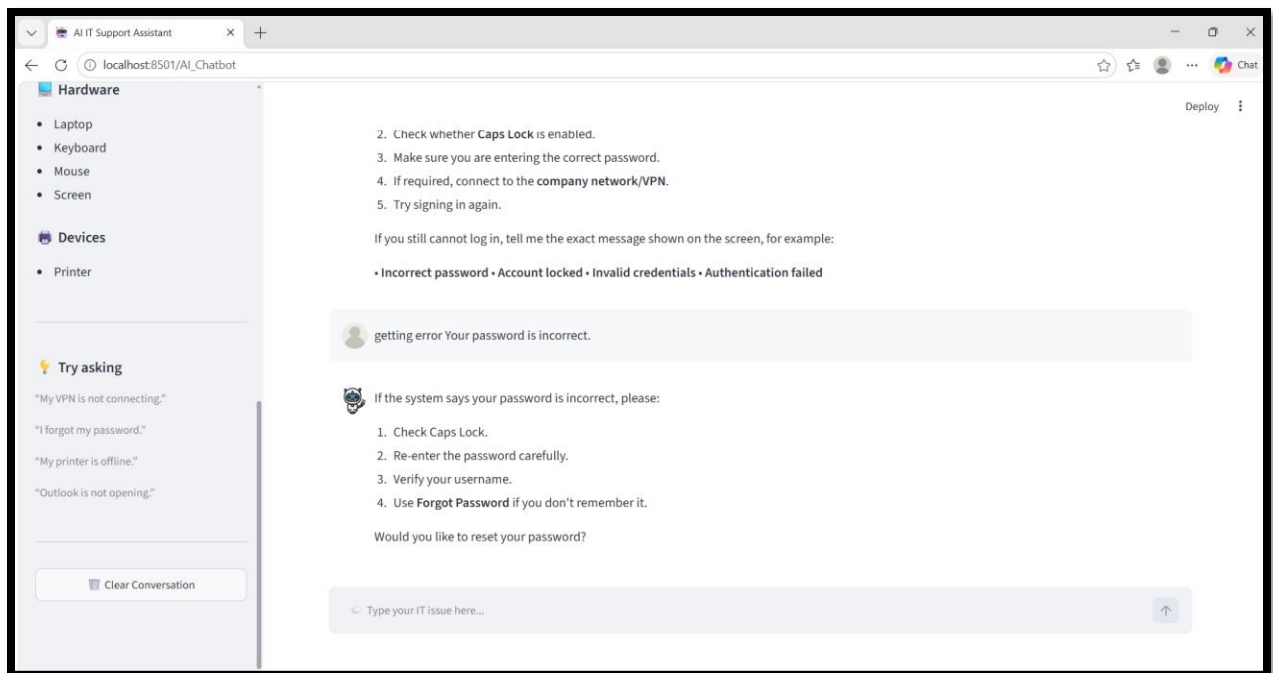
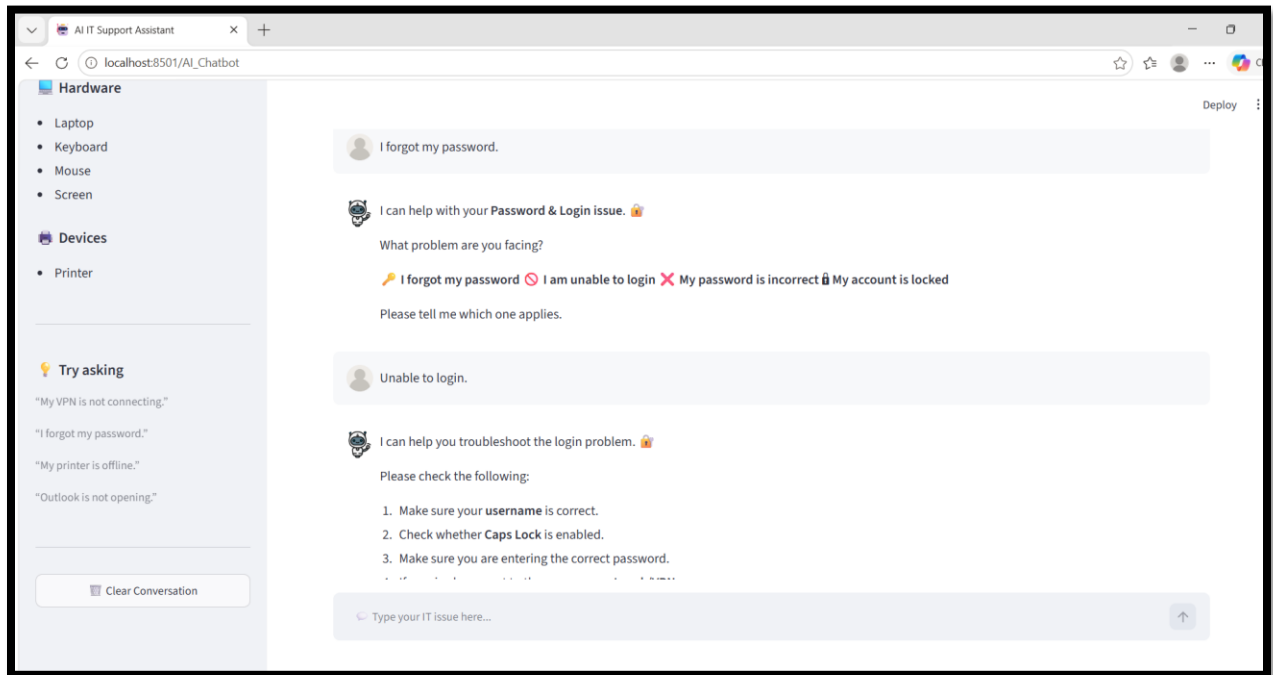
This screenshot shows the same web application after a prediction. The 'Enter Incident Description' field now contains 'VPN connection failed'. The 'Predict Category' button is disabled. Below the input field, a green box displays 'Predicted Category: Network'. A blue progress bar is shown below this. Under the heading 'Confidence', it states 'Prediction Confidence' followed by '90.0%'. At the bottom, a blue box with a globe icon says 'Suggested Team: Network Support'. The sidebar and 'Deploy' button remain the same.



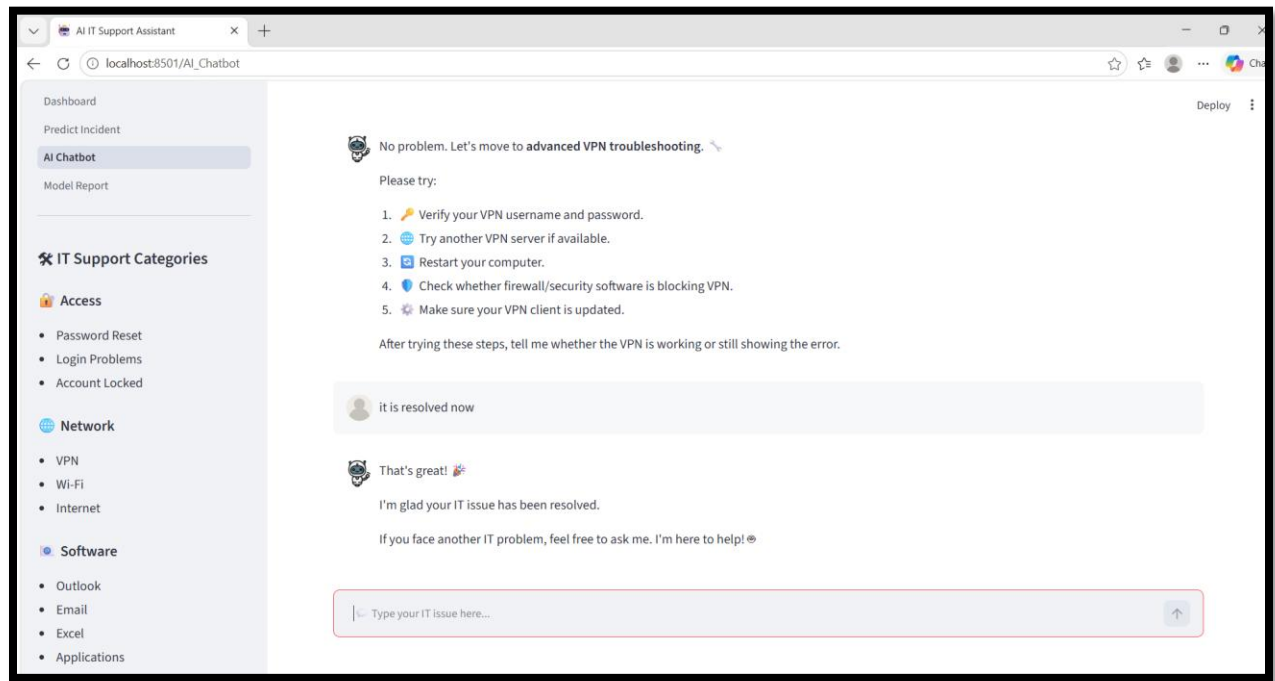
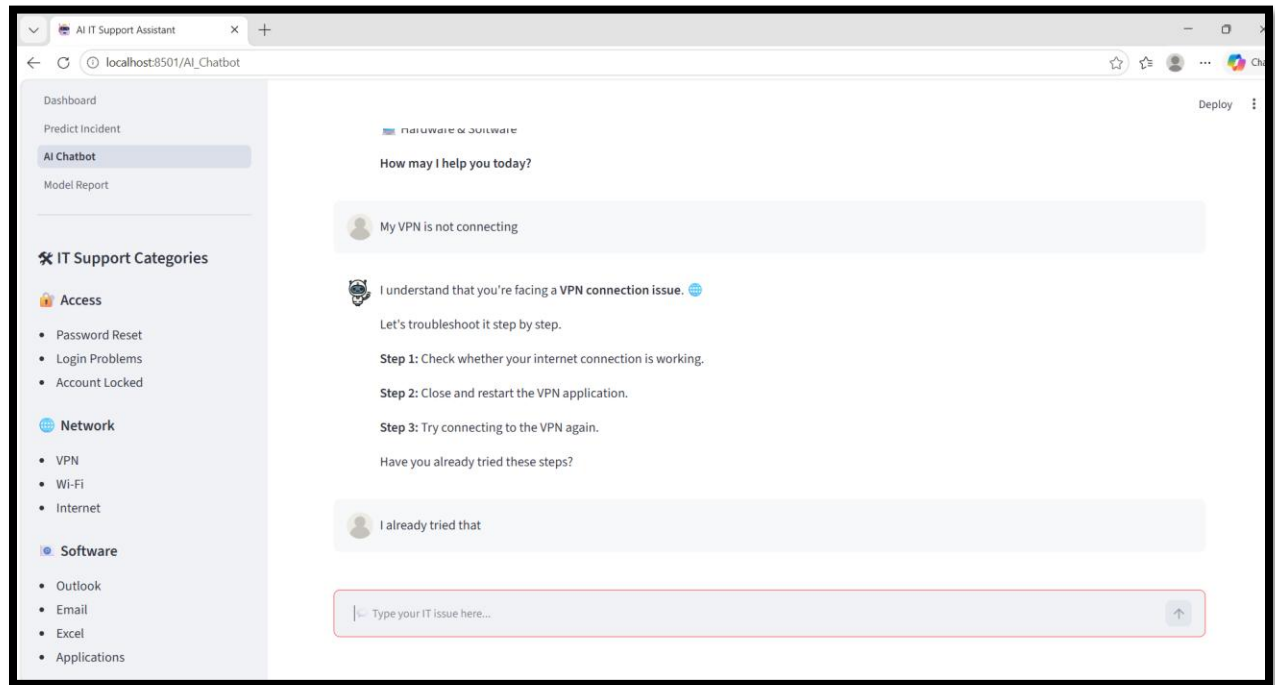
4) Chatbot Welcome Screen



5) Password Conversation



6) VPN Follow-up Conversation



7) Non-IT Question Handling

The screenshot shows a web application titled "AI IT Support Assistant" running on localhost8501/AI_Chatbot. The interface includes a sidebar with navigation links: app, Dashboard, Predict Incident, AI Chatbot (selected), and Model Report. Below the sidebar, there are "IT Support Categories" for Access, Network, and Software. The main content area displays a list of support topics: Password & Login, VPN / Network / Wi-Fi, Email & Outlook, Printer Issues, and Hardware & Software. A chatbot icon is present, and a message from the user asks, "What is the capital of France?". The chatbot responds with a message stating it is an AI IT Support Assistant and can only help with IT-related issues. It lists the categories it can assist with: Password & Login, VPN / Network / Wi-Fi, Email / Outlook, Printer, Hardware, and Software / Applications. An example message is provided: "Example: 'I forgot my password and cannot log in.'". A text input field at the bottom is labeled "Type your IT issue here...".

8) Model Report

